AI/ML Task 3 – Model Validation, Overfitting Control & Hyperparameter Tuning Report

1. Objective

The objective of this project is to improve the reliability and generalization of machine learning models for predicting California housing prices. The workflow focuses on model validation, detecting overfitting, applying cross-validation, performing hyperparameter tuning using GridSearchCV, and selecting the best model based on regression metrics.

2. Dataset & Features

Dataset: California Housing Dataset (scikit-learn)
• Total Samples: 20,640
• Input Features: 8 numerical features
• Target Variable: HousePrice (Median House Value)

Features Used:
• MedInc
• HouseAge
• AveRooms
• AveBedrms
• Population
• AveOccup
• Latitude
• Longitude

3. Data Preprocessing

The dataset was inspected for missing values and summary statistics. The independent variables (X) and target variable (y) were separated. StandardScaler was applied to standardize the feature values before model training. The dataset was then divided into 75% training data and 25% testing data using random_state = 42.

4. Baseline Models

The following regression models were trained as baseline models:
• Linear Regression
• Ridge Regression (alpha = 1.0)
• Decision Tree Regressor

5. Overfitting Analysis

The Decision Tree Regressor was evaluated using both training and testing RMSE. A large difference between training and testing performance indicates overfitting. This analysis helps determine whether the model generalizes well to unseen data.

6. Cross-Validation

Five-fold Cross-Validation (5-Fold CV) was performed using cross_val_score() to obtain a more reliable estimate of model performance. The average cross-validation RMSE was calculated for comparison across models.

7. Hyperparameter Tuning

GridSearchCV was used to optimize the Decision Tree Regressor by searching different combinations of max_depth, min_samples_split, and min_samples_leaf. The best parameter combination was selected based on cross-validation performance.

8. Model Evaluation

The optimized model was evaluated using:
• RMSE (Root Mean Squared Error)
• R^2 Score

Model comparison charts were also generated to compare baseline and tuned models.

9. Best Model Selection

Best Model: Tuned Decision Tree Regressor

Reasons:
• Reduced overfitting through optimized hyperparameters.
• Better balance between training and testing performance.
• Improved Cross-Validation results.
• Achieved competitive RMSE and R^2 Score on unseen data.

10. Performance Analysis

Cross-validation demonstrated that the tuned model provides more consistent performance across different data splits. Hyperparameter tuning improved the model's ability to generalize while reducing the risk of overfitting.

11. Model Saving

The final optimized Decision Tree model and the StandardScaler were saved using Joblib. Saving trained artifacts allows future predictions without retraining and supports deployment.

12. Conclusion

This project demonstrated a complete machine learning workflow involving preprocessing, model validation, overfitting detection, cross-validation, hyperparameter tuning, evaluation, visualization, and model persistence. The tuned Decision Tree Regressor was selected as the final model because it achieved the best balance between prediction accuracy and generalization.

13. Future Improvements

• Evaluate ensemble models such as Random Forest and XGBoost.
• Perform RandomizedSearchCV for faster tuning.
• Engineer additional features.
• Deploy the optimized model using Streamlit or Flask.
• Automate model retraining using ML pipelines.